Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass

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Abstract

We study multi-view 3D reconstruction from large, unordered, and unposed image sets. Existing feed-forward pointmap methods such as DUSt3R regress geometry from image pairs and then rely on quadratic pair enumeration and a separate global alignment stage, which makes latency and memory grow rapidly with the number of views. We present Fast3R, a transformer that processes all input images jointly and predicts per-view pointmaps for the whole set in a single forward pass. Fast3R concatenates patch tokens from all views, performs all-to-all attention over the full set, and regresses both a global pointmap in a shared anchor frame and a local pointmap in each camera frame, together with confidence maps. A random image-index embedding scheme lets the model train on short sequences while extrapolating to much larger test-time view counts. On multi-view camera pose estimation, Fast3R reaches 99.7% RRA@15 and 82.5 mAA(30) on 10-view CO3Dv2 with the reported no-outdoor variant, while running at 251.1 FPS. On RealEstate10K, the full model obtains 72.7 mAA(30). On scene reconstruction, Fast3R is competitive with DUSt3R and Spann3R on DTU, 7-Scenes, and Neural RGB-D while being substantially faster. On a single A100 at 512×384 *resolution, it processes 32 views in 0.509 seconds and scales to 1000 views in 137.62 seconds using 63.01 GiB, whereas DUSt3R runs out of memory beyond 48 views.*

1. Introduction

Multi-view 3D reconstruction is a core problem in vision, underpinning robotics, AR/VR, view synthesis, and embodied perception. Classical structure-from-motion and multi-view stereo pipelines recover camera poses and geometry through feature matching, bundle adjustment, and depth fusion [7, 17, 18, 29]. These systems remain powerful, but they depend on carefully engineered optimization stages and can struggle when calibration is unknown, overlap is sparse, or image collections are large and unordered.

Recent feed-forward alternatives have made 3D prediction more accessible by directly regressing pointmaps from images [8, 23, 26, 28]. In particular, DUSt3R [26] showed that predicting dense 3D points in image coordinates is a useful representation that unifies reconstruction, depth, and pose recovery. However, DUSt3R remains fundamentally pairwise: for an N-image collection it requires $O(N^2)$ pair processing, followed by a separate alignment procedure. This pairwise bottleneck becomes the dominant limitation at scale, both computationally and statistically. As more views are added, inference cost and memory rise quickly, and inconsistencies introduced at the pair level must later be repaired by global optimization.

This paper asks whether large-scale reconstruction can instead be framed as a *single* multi-view prediction problem. Our key observation is that the ambiguity faced by any one view often becomes resolvable when the entire image set is available simultaneously. A model that attends jointly over all views can use long-range geometric context directly, without first committing to pairwise predictions. This motivates Fast3R, a transformer architecture that takes an unordered set of images and predicts pointmaps for all of them in one forward pass.

Fast3R uses a CroCo-initialized ViT-L style image encoder [27], concatenates patch tokens from all views, and applies a multi-view decoder with global self-attention [22]. The model outputs two dense predictions per image: a global pointmap expressed in a common anchor frame and a local pointmap expressed in the image's own camera frame. We supervise both heads with normalized 3D regression and learned confidences. To bridge the gap between the 20-view training regime used in our main configuration and much larger test sets, we inject a 1D image-index embedding and randomly sample non-repeating view indices during training, which encourages robustness to sparse placements within a much larger positional range.

Empirically, Fast3R changes the efficiency frontier for pose-free multi-view reconstruction. The method is competitive with strong pointmap baselines on reconstruction benchmarks and offers much better throughput. On camera pose estimation, the provided results show very strong accuracy on CO3Dv2 and strong generalization to RealEstate10K. On system scaling, Fast3R handles 1000 views in one forward pass on a single A100 GPU, while the pairwise DUSt3R baseline runs out of memory once the number of views exceeds 48.

Our contributions are:

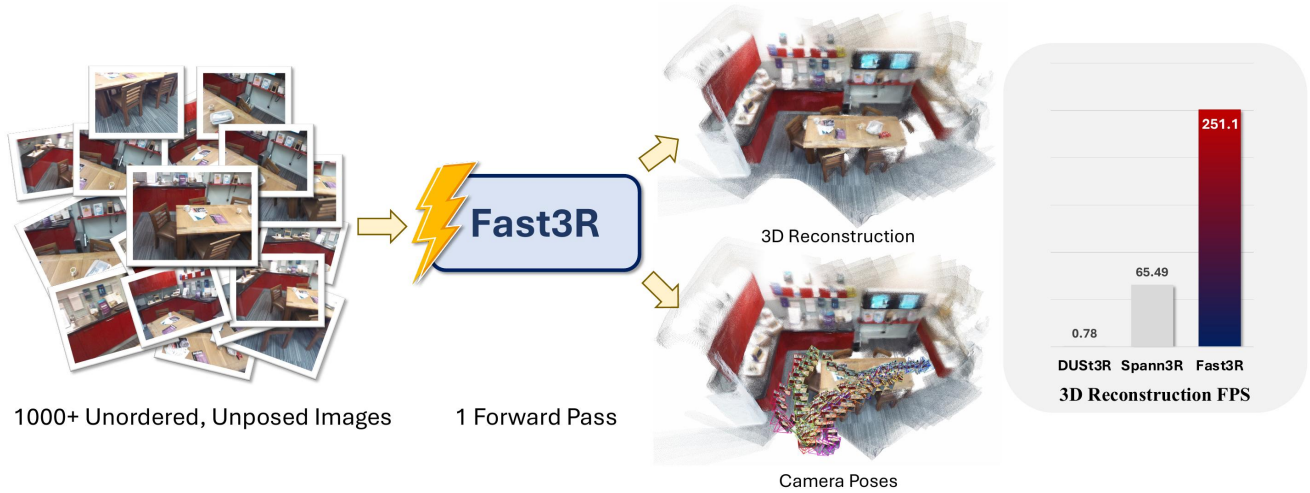

Figure 1. **Fast3R is a method towards 3D reconstructing 1000+ unordered, unposed images in a single forward pass.** Instead of enumerating image pairs and aligning them afterward, Fast3R jointly reasons over all views and directly predicts per-view pointmaps with confidences.

- A single-pass multi-view pointmap regressor that replaces pairwise reconstruction and post-hoc alignment with joint reasoning over the full image set.
- A multi-view transformer design with image-index embeddings and random index sampling that supports extrapolation from short training sequences to much larger test-time view counts.
- A dual-head formulation that predicts both global and local pointmaps with confidence-weighted supervision, enabling accurate downstream reconstruction after local-to-global alignment.
- An empirical study showing competitive reconstruction quality, strong pose estimation accuracy, and orders-of-magnitude improvements in throughput and scalability.

2. Related Work

2.1. Classical Reconstruction Pipelines

Classical SfM and MVS systems estimate camera poses from sparse correspondences and then recover dense geometry through multi-view consistency [7, 17, 18]. These methods remain strong on well-conditioned image sets and are still widely used in practice, but their optimization-heavy structure makes them brittle when calibration is missing or overlap is weak. Learned SLAM systems such as DROID-SLAM [21] and ORB-SLAM variants [12, 13] improve robustness in sequential settings, yet they still rely on explicit geometric optimization and are not designed for large unordered collections.

2.2. Learned Pointmap and Pose-free Reconstruction

DUSt3R [25, 26] introduced pointmap regression as a versatile representation for unconstrained 3D vision, showing that camera parameters can be recovered after predicting dense 3D points. MASt3R [8] extends this family with local features for image matching, while Spann3R [23] predicts globally expressed pointmaps for ordered or unordered sequences using an external memory. FrozenRecon [28] takes a different route and combines frozen monocular depth models with test-time optimization. These methods substantially lower the barrier to 3D reconstruction, but they still inherit limitations from pairwise processing, sequential memory updates, or iterative test-time refinement. Fast3R targets the remaining gap: a feed-forward architecture that reasons over all views jointly and scales gracefully to very large image sets.

2.3. Multi-view Pose Estimation

Sparse-view pose estimation has also seen a shift toward set-level learning. Feature-matching pipelines based on SuperPoint and SuperGlue [6, 15] and refinement systems such as PixSfM [11] remain strong baselines. Learning-based multi-view pose predictors include RelPose [32], PoseDiffusion [24], and RelPose++ [10]. These approaches mostly regress or sample camera parameters directly. In contrast, Fast3R predicts dense geometry first and recovers camera poses from its pointmaps, following the geometry-first philosophy of DUSt3R while removing the pairwise bottleneck.

3. Method

3.1. Problem Setup

We consider an unordered collection of N RGB images of a static scene or object. Neither camera intrinsics nor camera poses are given at inference time. The goal is to predict a dense 3D pointmap for each image and use these predictions for downstream camera pose estimation and 3D reconstruction.

Following DUSt3R [26], we represent geometry with a per-pixel pointmap rather than with depth alone. This representation is convenient because it can express geometry in either a shared global frame or an image-local frame, and it supports downstream camera recovery via PnP and point cloud fusion.

3.2. Joint Multi-view Transformer

Figure 2 summarizes the architecture. Each image is first encoded independently by a CroCo-style vision transformer encoder [27] with patch size 16, 24 encoder blocks, and 1024-dimensional tokens in the main model configuration. The encoder uses rotary position embeddings [20] over 2D patch coordinates.

The encoded patch tokens from all views are then concatenated into a single long sequence. A transformer decoder processes this entire sequence with dense self-attention, so each patch can attend to all patches from all images. This removes the need to choose image pairs or to propagate information through a sequential memory. In the main training configuration, the decoder is also 24 layers deep with 1024-dimensional hidden states and FlashAttention-enabled attention kernels [4, 5] for efficiency.

The decoder receives two kinds of position information. First, each patch retains its 2D image position. Second, each patch receives an image-index embedding that identifies which input image it came from. Rather than assigning consecutive image IDs during training, Fast3R samples non-repeating indices from a larger bank of sinusoidal embeddings. The anchor image keeps index 0, while the remaining views are assigned random indices. This design exposes the model to sparse placements in a larger positional range and improves generalization when the number of test views far exceeds the number seen during training.

3.3. Dual Pointmap Heads

Fast3R predicts two dense outputs for each image:

1. a **global head** that predicts a pointmap in a common anchor-centered frame, and
2. a **local head** that predicts a pointmap in that image's own camera frame.

Both outputs are produced by DPT-style dense prediction heads [26]. Each head also predicts a confidence map. The global head is used directly for pose estimation and provides a shared coordinate system across views. The local head is more flexible for per-view geometry because it does not need to express all points in a single anchor frame. At evaluation time, we align the local prediction of each image to the global prediction using a confidence-filtered rigid registration with scale, yielding a fused representation that improves reconstruction quality in scene benchmarks.

3.4. Training Objective

Training uses multi-view pointmap regression in both coordinate systems. For the global head, all ground-truth pointmaps are transformed into the anchor view's camera frame. For the local head, each ground-truth pointmap is transformed into its own camera frame. The code implements a normalized $L_{2,1}$ regression loss with average-distance normalization across valid 3D points, which reduces sensitivity to overall scale.

The confidence maps are trained jointly with the geometry through a confidence-weighted objective similar to DUSt3R. Intuitively, large residuals are penalized more when the model claims high confidence, while uncertain regions can be down-weighted at the cost of a log-confidence penalty. The final training criterion averages global and local confidence-weighted regression terms. In the released configuration, the confidence weight uses $\alpha = 0.2$.

3.5. Implementation for Large-view Inference

Fast3R is designed to keep the whole image set in a single forward graph, but practical scaling still requires careful implementation. The encoder can process images in chunks to avoid out-of-memory failures, and the dense heads are applied to subsets of views when necessary. FlashAttention reduces the cost of the multi-view decoder, and the model supports mixed-resolution training. These choices make it possible to scale from short training sequences to hundreds or thousands of views at test time on a single A100 GPU.

4. Experiments

4.1. Training and Evaluation Setup

The main training configuration uses 20 input views per sample and validates on 10-view samples. Training data is drawn from six multi-view sources: CO3D [14], ScanNet++ [31], ARKitScenes [3], Habitat [16], BlendedMVS [30], and MegaDepth [9]. The model is initialized from a DUSt3R ViT-L checkpoint and then trained end-to-end with the multi-view objective described above.

We report two groups of tasks. For **camera pose estimation**, we evaluate on 10-view CO3Dv2 windows and on RealEstate10K [33]. For **reconstruction**, we evaluate on DTU [1], 7-Scenes [19], and Neural RGB-D [2]. We use the provided benchmark tables and figures directly. For scene

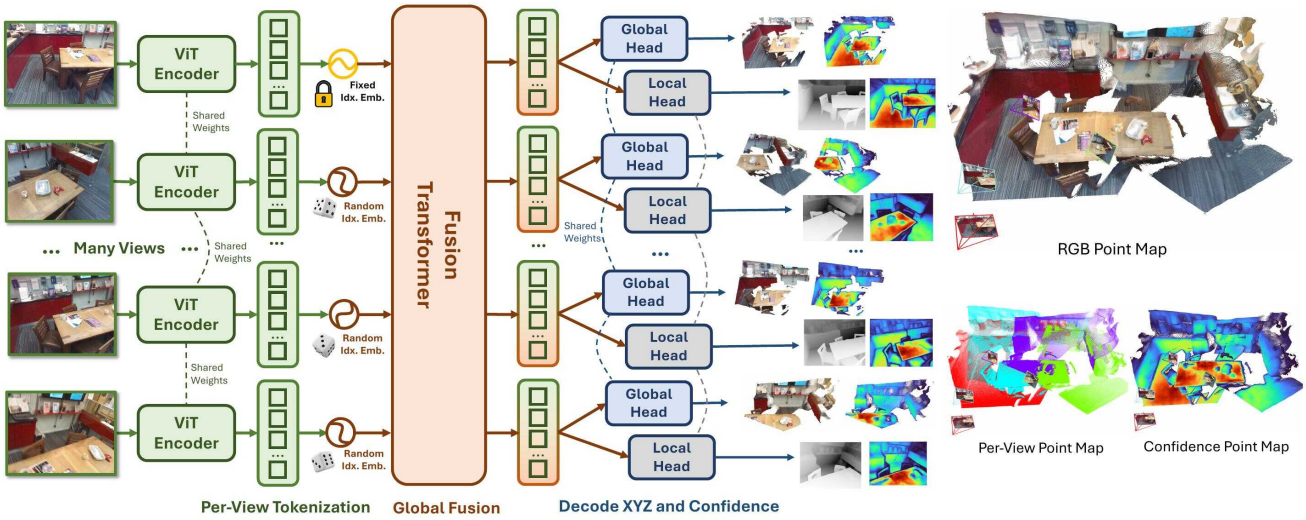

Figure 2. **Model architecture of Fast3R.** Each image is encoded independently into patch tokens, tokens from all views are concatenated, and a multi-view transformer performs all-to-all attention over the full set. Two DPT heads regress a global pointmap in a shared anchor frame and a local pointmap in each view's frame, together with confidence maps.

reconstruction, distances are reported as median point-to-point errors; for pose estimation, we report relative rotation accuracy (RRA), relative translation accuracy (RTA), and mean average accuracy mAA(30).

Unless stated otherwise, pose metrics are computed after recovering cameras from the predicted pointmaps, not from a separate pose-regression head. This keeps the comparison aligned with the geometry-first design of DUSt3R-style methods and makes the accuracy numbers reflect cross-view geometric consistency rather than a different downstream parameterization.

4.2. Multi-view Pose Estimation

Table 1 compares Fast3R against classical matching pipelines and recent learned pose estimators. The reported **Fast3R-no-outdoor** variant reaches the strongest CO3Dv2 rotation metrics in the table, with 99.7% RRA@15 and 97.4% RRA@5, and also achieves the best mAA(30) at 82.5. The full Fast3R checkpoint is slightly weaker on CO3Dv2, obtaining 75.0 mAA(30), but remains highly competitive and substantially faster than all compared methods. On RealEstate10K, the full model reaches 72.7 mAA(30), outperforming DUSt3R and trailing MASt3R.

The efficiency gap is especially large. Fast3R runs at 251.1 FPS in the reported setting, versus 0.78 FPS for DUSt3R and 0.23 FPS for MASt3R. This supports the main motivation of the paper: removing pairwise enumeration changes the practical operating regime of dense geometry-based pose estimation.

4.3. 3D Reconstruction

Tables 2 and 3 show that Fast3R remains competitive on reconstruction quality while delivering much higher speed. On 7-Scenes, Fast3R obtains 1.58 Acc and 0.93 Comp, compared with 1.23 and 0.91 for DUSt3R† and 1.48 and 0.85 for Spann3R. On Neural RGB-D, Fast3R reaches 3.40 Acc and the best completeness at 1.01. On DTU, Fast3R obtains 1.706 Acc and the best completeness at 0.857. These results indicate a consistent pattern: the single-pass formulation does not always win on every metric, but it preserves competitive geometry quality while shifting the runtime-quality tradeoff decisively in its favor.

4.4. System Scaling

Table 4 isolates the systems aspect of the method. At 512×384 resolution on a single A100, Fast3R processes 32 views in 0.509 seconds with 13.25 GiB peak memory, whereas DUSt3R requires 129.0 seconds and 67.61 GiB. DUSt3R runs out of memory at 48 views because all pairwise reconstructions must still be globally aligned. In contrast, Fast3R scales to 320, 800, 1000, and even 1500 views in one forward pass, reaching 137.62 seconds and 63.01 GiB at 1000 views. This is the clearest evidence that the architectural change from pairwise to joint reasoning materially improves scalability.

Methods	CO3Dv2 [14]					RealEstate10K [33]	FPS
	RRA@15↑	RRA@5↑	RTA@15↑	RTA@5↑	mAA(30)↑	mAA(30)↑	
Colmap+SG [6, 15]	36.1	24.4	27.3	17.2	25.3	45.2	0.056
PixSfM [11]	33.7	26.1	32.9	17.6	30.1	49.4	-
RelPose [32]	57.1	-	-	-	-	-	0.02
PosReg [24]	53.2	-	49.1	-	45.0	-	0.015
PoseDiff [24]	80.5	59.5	79.8	61.7	66.5	48.0	0.015
RelPose++ [10]	(85.5)	-	-	-	-	-	0.02
DUSt3R [25]	96.2	-	86.8	-	76.7	67.7	0.78
MASt3R [8]	94.6	93.2	**91.9**	**86.2**	81.8	**76.4**	0.23
Fast3R-no-outdoor (Ours)	**99.7**	**97.4**	87.1	76.1	**82.5**	-	**251.1**
Fast3R (Ours)	96.2	90.2	81.6	68.2	75.0	72.7	**251.1**

Table 1. **Multi-view pose regression on the CO3D [14] and RealEstate10K [33] datasets**. Parentheses denote methods that do not report results on the 10 views set; we report their best for comparison (8 views). Fast3R does not assume known camera intrinsics.

Method	FPS	7 scenes [19]		NRGBD [2]	
		Acc↓	Comp↓	Acc↓	Comp↓
F-Recon [28]	<0.1	7.62	2.31	20.59	6.31
DUSt3R† [26]	0.78	**1.23**	0.91	**2.51**	1.03
Spann3R [23]	65.4	1.48	**0.85**	3.15	1.10
Fast3R (Ours)	**251.1**	1.58	0.93	3.40	**1.01**

Table 2. **Quantitative reconstruction results on scene datasets:** The numbers indicate `median` distance to GT points on 7-Scenes [19] and NRGBD [2] datasets. These datasets contain video trajectories of 500-1500 frames, and we evaluate using the same frame skip as other baselines. For 7-Scenes this is `skip`=20, and NRGBD uses `skip`=40. DUSt3R† indicates using DUSt3R's final weights on 224 × 224 images, to fit within GPU memory. Distances are scaled 100× to remove the leading `0.00`.

Method	Views	Acc↓	Comp↓
		Med.	Med.
DUSt3R [26]	all/5	**1.159**	0.914
DUSt3R† [26]	all/5	1.297	1.002
Spann3R [23]	all/5	2.268	1.295
Fast3R (Ours)	all/5	1.706	**0.857**

Table 3. **Quantitative results on object-centric DTU [1] dataset.** Using a `skip`=5 on trajectories of 49 frames.

# Views	**Fast3R**		**DUSt3R**	
	Time (s)	Peak GPU Mem (GiB)	Time (s)	Peak GPU Mem (GiB)
2	0.065	3.84	0.092	3.52
8	0.122	6.33	8.386	24.59
32	0.509	13.25	129.0	67.61
48	0.84	20.8	OOM	OOM
320	15.938	41.90	OOM	OOM
800	89.569	55.97	OOM	OOM
1000	137.62	63.01	OOM	OOM
1500	308.85	78.59	OOM	OOM

Table 4. **System performance metrics for different view counts on Fast3R and DUSt3R on a single A100.** Time is measured in seconds (s), and memory is measured in gibibytes (GiB). Each view is 512x384 in resolution. For DUSt3R, at 48 views the N^2 pairwise reconstructions eventually consume all VRAM at its global alignment stage. Note that Fast3R's reported fastest FPS of 251.1 uses 108 views in 224x224 resolution.

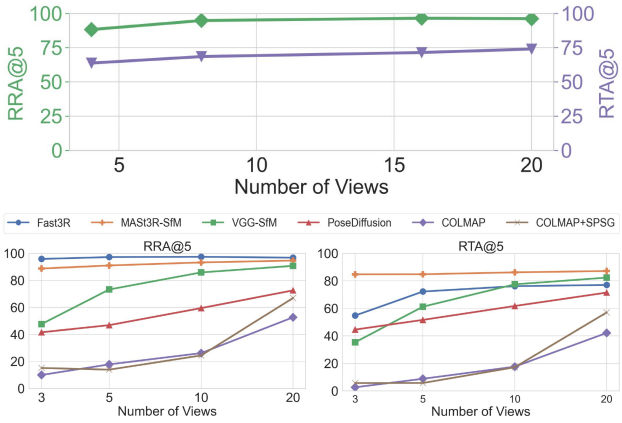

Figure 3. **Pose estimation improves as the view budget grows.** Top: training with more views improves CO3D pose estimation. Bottom: test-time pose accuracy also rises when more views are provided to the same model.

5. Analysis

5.1. More Views Help When the Model Can Use Them Jointly

Figures 3 and 4 summarize a central property of Fast3R: its accuracy improves when more views are available. This is not automatic for pairwise methods because extra views mainly increase the number of pair evaluations and the complexity of global alignment. In Fast3R, additional images directly enlarge the decoder's context, giving each patch access to more disambiguating evidence. The supplied figures show gains in pose estimation with more training views and more test views, and similar improvements for scene reconstruction.

The benefit is not uniform across benchmarks, which is itself informative. On long indoor trajectories such as 7-Scenes and Neural RGB-D, extra context improves both coverage and stability because distant revisits can still exchange information through the same attention graph. On DTU, the gain is smaller and sometimes non-monotonic,

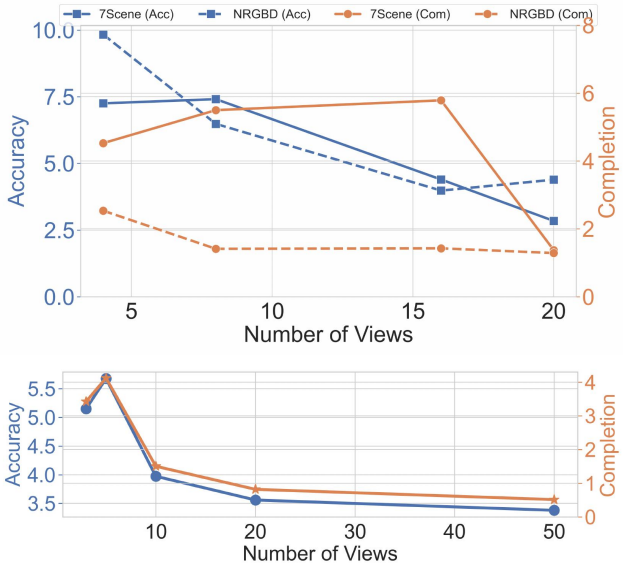

Figure 4. **Reconstruction benefits from additional context as well.** Top: larger training-view budgets improve reconstruction on 7-Scenes and Neural RGB-D. Bottom: DTU quality varies with the number of test views, but the model remains usable far beyond the short training regime.

Pointmap Type	7-Scenes [19]		NRGBD [2]		DTU	
	Acc↓	Comp↓	Acc↓	Comp↓	Acc↓	Comp↓
local aligned to global	2.84	1.37	4.39	1.28	3.91	1.41
global	4.81	1.64	4.85	1.32	3.88	1.41
Δ	+1.97	+0.27	+0.46	+0.04	-0.03	0.00

Table 5. **Ablation on the effect of local head on 3D reconstruction.** Red/green indicate an increase/decrease in error compared to using the local pointmap aligned to the global pointmap.

suggesting that once an object-centric capture already contains strong overlap, further views mainly help completeness rather than fine local accuracy. This pattern is consistent with the quantitative tables: the single-pass design is most valuable when global scene context is genuinely hard to aggregate with pairwise reasoning.

5.2. The Local Head Matters for Reconstruction

Table 5 isolates the contribution of the local head. Replacing the aligned local output with the raw global output substantially worsens scene reconstruction on 7-Scenes and Neural RGB-D: on 7-Scenes, Acc degrades from 2.84 to 4.81 and Comp from 1.37 to 1.64; on Neural RGB-D, Acc degrades from 4.39 to 4.85 and Comp from 1.28 to 1.32. On DTU, the difference is negligible. This suggests that for long indoor scene trajectories, predicting geometry in each view's own frame and only then aligning it to the global frame is more stable than forcing all dense predictions to live directly in the anchor frame.

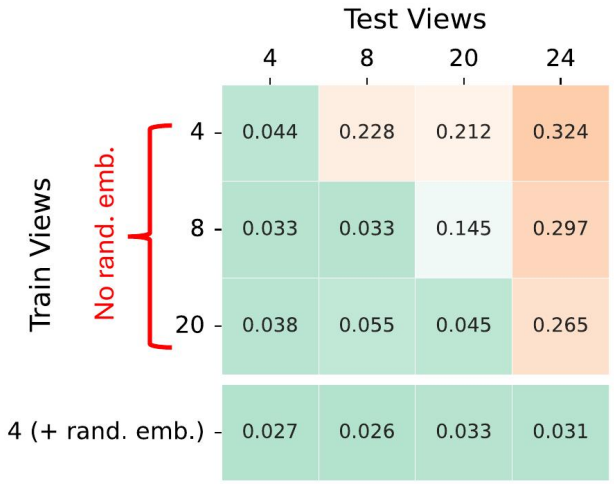

Figure 5. **Effect of sampling image-index positional embeddings during training.** This ablation evaluates the image-index randomization scheme used by Fast3R to support train-short, test-long behavior.

5.3. Image-index Randomization Improves Extrapolation

The random image-index embedding scheme is a simple but important part of the design. Without it, a model trained on 20 views only sees a narrow and contiguous range of image indices. Sampling non-repeating indices from a much larger embedding bank makes the decoder less tied to a particular training-time view count. Figure 5 reports this ablation directly.

This mechanism can be viewed as regularization against slot overfitting. If training always uses consecutive IDs, the decoder can partially memorize ordinal patterns that do not survive when hundreds of views are presented at test time. Sparse randomized indices instead encourage attention to rely on image content and multi-view agreement, which is exactly the behavior needed for train-short, test-long generalization.

5.4. Qualitative Results

Figure 6 shows representative outputs. Qualitatively, Fast3R reconstructs coherent geometry across wide-baseline image collections and long scene trajectories. The most noticeable artifacts appear in thin structures, low-texture regions, and views with little overlap, which is consistent with the remaining quantitative gap to the strongest reconstruction baselines on some accuracy metrics.

6. Conclusion

Fast3R reformulates multi-view 3D reconstruction as a single forward pass over the entire image set. By replac-

Figure 6. **Qualitative examples of Fast3R's output.** The predicted pointmaps are coherent across large image sets and support both object-centric and scene-level reconstructions without an explicit global alignment stage at inference time.

ing pairwise processing and post-hoc alignment with joint multi-view attention, the method scales to far larger numbers of views while remaining competitive on reconstruction quality and strong on pose estimation. The results support a clear conclusion: for large unordered image collections, end-to-end global reasoning is a more scalable operating point than pairwise geometry aggregation.

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